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Metabolomic Insights Into Saliva as a Non-Invasive Surrogate for Serum to Enable Community-Based Biomarker Investigation

Alireza Abdshah^{1,2} | Mohammad Alyamani^{1,2} | Robert A. Burgess^{1,2} | Dipen J. Parekh^{1,2} | Benjamin Spieler³ | Rakesh Singal² | Matthew C. Abramowitz³ | Jaime R. Merchan^{2,4} | Lucia Pannunzio^{1,2} | Brandon A. Mahal³ | Erin Kobetz^{2,5,6} | Sanj Punnen^{1,2} | Marijo Bilusic² | Nima Sharifi^{1,2} 

¹Desai Sethi Urology Institute, University of Miami Miller School of Medicine, Miami, Florida, USA | ²Sylvester Comprehensive Cancer Center, University of Miami Miller School of Medicine, Miami, Florida, USA | ³Department of Radiation Oncology, University of Miami Miller School of Medicine, Miami, Florida, USA | ⁴Department of Medicine, Medical Oncology Division, University of Miami Miller School of Medicine, Miami, Florida, USA | ⁵Department of Public Health Sciences, University of Miami Miller School of Medicine, Miami, Florida, USA | ⁶Department of Medicine, University of Miami Miller School of Medicine, Miami, Florida, USA

Correspondence: Nima Sharifi (nimasharifi@miami.edu)

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ABSTRACT

Introduction: Using serum metabolomics in Community-Based Participatory Research (CBPR) with underserved populations presents challenges due to the invasive nature of venipuncture, which limits longitudinal sample collection in community settings. In this pilot study, we compared saliva and serum metabolomics in patients with prostate cancer to evaluate saliva's potential in biomarker research and identified metabolic changes associated with androgen deprivation therapy (ADT).

Methods: Blood and saliva were collected from 20 eugonadal patients with prostate cancer and 20 patients receiving ADT. Untargeted metabolomics of metabolites was performed, and glucocorticoids were quantified using targeted metabolomics. Data were analyzed using MassHunter, METLIN database, and MetaboAnalyst.

Results: Across all samples, 1198 metabolites were identified. Comparing ADT to eugonadal serum and integrating fold change ($FC > 2$ or $FC < 0.5$) and significance (adjusted $p < 0.1$), we identified 14 upregulated and 20 downregulated metabolites. Comparing ADT to eugonadal saliva, we identified 14 upregulated and 28 downregulated metabolites. Targeted metabolomics confirmed correlations between serum and salivary cortisol and salivary and serum cortisone ($r = 0.61$ and 0.62 , $p < 0.001$), underscoring the potential of saliva glucocorticoids as biomarkers for stress-related metabolic alterations and suggesting that saliva reflects the systemic metabolomic profile.

Conclusion: ADT-associated metabolomic changes are detectable in saliva. Targeted analysis suggests that salivary glucocorticoids may serve as biomarkers of metabolism and stress in community-based studies. Our findings support saliva as a non-invasive alternative to serum for biomarker research, facilitating community-based investigations and advancing public health efforts.

Alireza Abdshah and Mohammad Alyamani contributed equally and are co-first authors.

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1 | Introduction

Metabolomics demarcates biological phenotypes by profiling changes in endogenous small molecules. It has proven to be valuable in translational prostate cancer research, having led to the identification of biomarkers for prediction, diagnosis, progression, prognosis, and recurrence [1–3]. The most common biological specimens used in metabolomic studies are body fluids (blood, plasma, serum, and urine) [4]. Blood collection requires regular visits to the clinic and performance by qualified and trained personnel. The process is burdensome for patients, especially those who are not easily able to travel to the clinic, as it often requires time off work and transportation costs. For people with chronic conditions or limited mobility, these challenges can create considerable barriers to consistent and effective monitoring and participation in studies, negatively impacting overall inclusivity and accurate representation [5]. Therefore, noninvasive sample collection would greatly benefit patients with limited access to medical care, as well as simply being more convenient for all patients. Saliva has been considered as a biofluid for metabolomics analysis because collection is noninvasive, and sample handling is simple. Also, saliva, due to its dynamic composition as influenced by local and systemic factors, reflects systemic physiological changes, and therefore is promising for non-invasive metabolic profiling. Moreover, saliva has been reported to reflect the serum levels of some metabolites [6–10].

Androgen deprivation therapy (ADT), as a cornerstone treatment for advanced prostate cancer, has a significant impact on systemic metabolism owing to suppression of androgen signaling. ADT induces profound alterations in body composition, including an increase in body fat mass and reduction in lean body mass, leading to metabolic dysregulation [11]. These effects originate from alterations in lipid and glucose metabolic pathways, potentially leading to an elevated risk of cardiovascular events [12]. Studies have consistently shown disrupted lipid profiles, altered amino acid metabolism, and dysregulated steroid hormone synthesis in ADT-treated patients [13–15]. These changes reflect both direct effects of androgen suppression and downstream metabolic adaptations.

Serum metabolomic analysis has emerged as a pivotal tool for investigating ADT-induced changes in metabolism. Untargeted metabolomic analysis provides a comprehensive approach to detecting and understanding these systemic effects by mapping alterations across diverse metabolic pathways and phenotypes, offering critical insights into the biological and metabolic consequences of ADT. Untargeted serum metabolomics in prostate cancer patients treated with ADT has revealed disruptions in pathways associated with inflammation, oxidative stress, and energy metabolism [16], emphasizing the need for integrated and consistent strategies to mitigate long-term health risks in this vulnerable population.

ADT may impact salivary metabolomics through metabolite alterations in circulation, as well as changes in local enzyme activity and androgen receptor activity, as the androgen receptor is expressed in normal salivary gland tissue [17, 18]. AR is also expressed in malignant salivary gland tissue, and hormonal therapies have activity in salivary gland cancers [19]. The 11 β -hydroxysteroid dehydrogenase-2 enzyme activity in salivary glands modulates glucocorticoid metabolism [20], suggesting

that saliva profiles might be influenced by ADT-induced endocrine changes, thus potentially enabling broader biomarker discovery for treatment monitoring.

There is a need for non-invasive, accessible methods for both cancer screening and monitoring over time after diagnosis [21–24]. Saliva sampling provides a rapid, cost-effective method for large-scale data collection in translational research and clinical monitoring, particularly in community health settings with persistent healthcare disparities. Glucocorticoids, key biomarkers of stress and metabolic regulation, are especially relevant in prostate cancer due to their association with ADT [25], chronic disease risk, and socioeconomic status and health disparities [26, 27]. Integrating salivary metabolomics into public health efforts could facilitate early detection and personalized treatment strategies in prostate cancer, other cancers, and chronic inflammatory and stress-related conditions.

Comparing serum and salivary metabolomics offers a unique opportunity to enhance our understanding of metabolic changes induced by hormonal treatment for prostate cancer and to develop sustainable, community-based health programs. While serum remains the gold standard for systemic metabolic profiling, saliva's ability to reflect systemic changes and its non-invasive collection make it a valuable surrogate for biomarker studies. Integrating saliva-based metabolomics into catchment area-relevant research could help bridge healthcare accessibility gaps, support community screening efforts, enable earlier interventions, and advance personalized therapies for under-represented populations. Therefore, we conducted a pilot study to determine whether metabolomic analysis of saliva is a feasible surrogate for serum metabolomics.

2 | Methods

2.1 | Patient Selection and Sample Handling

This study used samples collected from two cohorts of prostate cancer patients. In the first cohort, 20 men with prostate cancer but without prior hormonal therapy (eugonadal), and in the second cohort, 20 men with prostate cancer treated with hormonal therapy were recruited. All participants provided appropriate prior written informed consent for participation in the study. Blood and saliva were collected from each patient during their routine visits to the Desai Sethi Urology Institute and Sylvester Comprehensive Cancer Center. Samples were handled per our University of Miami Health System-approved IRB protocol (20230472). Blood was collected in red top Vacutainer Plus plastic serum collection tubes (BD367814, Becton Dickinson) and allowed to clot. The samples were centrifuged at 1430g for 10 min at room temperature within 1 h of collection. Stimulated saliva was collected by asking patients to chew a cotton swab that was then placed in a Salivette tube. Serum and saliva aliquots were frozen at -80°C until processing for analysis.

As an exploratory metabolomics study, a sample size of 20 participants per group was selected following recommendations for pilot investigations, which suggest that 12–20 subjects per group are sufficient to provide preliminary estimates of variability and effect size [28, 29]. All patients in the androgen deprivation therapy group were actively receiving hormonal

therapy at the time of sample collection. Because this study was cross-sectional and included patients referred from outside institutions, detailed information regarding duration of androgen deprivation therapy and prior systemic treatments was not uniformly available for all participants.

2.2 | Untargeted Metabolomics

All reagents used were LC-MS grade. The analysis was performed on an Agilent 6546Q time of flight instrument coupled with a 1290 Infinity II Bio UPLC system. Patient samples were processed following the Matyash liquid extraction protocol with cold methanol and tert-butyl methyl ether (MTBE), with slight modifications [30]: To 50 μ L serum or saliva, 225 μ L ice-cold methanol (Fisher) containing the internal standards (Metabolomics QReSS kit, Cambridge Isotope Laboratories Inc.) were added to precipitate proteins. Samples were then briefly vortexed and 750 μ L ice-cold MTBE (ACROS, Fair Lawn, NJ) added. Samples were vortexed for 2 min, then 188 μ L H₂O (Fisher) was added to achieve phase separation. The upper phase contains the non-polar metabolites plus lipids and was collected and processed independently from the lower phase, which contains the polar and semi-polar metabolites. After evaporation under nitrogen, the non-polar portion was reconstituted with a mixture of acetonitrile:methanol:water (25:25:50) and analyzed by reverse phase LC in positive polarity mode using a C18 Agilent Zorbax Eclipse (150 mm $\times$ 2.1 mm, 3.5 μ M particle size) analytical column at 40°C. Mobile phase A was 5 mM ammonium formate plus 0.1% formic acid in water; mobile phase B was 0.1% formic acid in acetonitrile:methanol:isopropanol (40:40:20), at a flow rate of 0.3 mL/min.

Polar metabolites were analyzed as follows. After evaporating the lower phase under nitrogen at 4°C, the samples were reconstituted with a mixture of acetonitrile:water (80:20). The analysis was carried out in positive and negative polarity mode using a hydrophilic interaction LC protocol. The metabolites were separated on an Agilent InfinityLab Poroshell 120 HILIC-Z analytical column (150 mm $\times$ 2.1 mm, particle size 2.7 μ m) at 15°C. The separation was achieved with 20 mM ammonium acetate (mobile phase A) and pure acetonitrile (mobile phase B).

2.3 | Targeted Metabolomics

To obtain accurate glucocorticoid levels, serum and saliva samples were subjected to LC-MS analysis following our previously published protocol [31]. To 200 μ L serum or saliva, 20 μ L internal standard (cortisol-d4 dissolved in H₂O: methanol; 1:1) was added followed by brief vortexing. Metabolites were extracted twice by adding 2 mL MTBE, vortexing for 3 min and then centrifuging for 5 min at 1430g at 4°C. The organic layer was then evaporated under nitrogen, and the samples were reconstituted in 200 μ L H₂O: methanol, 1:1. The LC-MS analysis was performed on our 6500+ mass analyzer (ABSciex) coupled to a Nexera-X3 UPLC system (Shimadzu). Metabolite separation was achieved by injecting 10 μ L sample on a C18 Agilent Zorbax Eclipse analytical column (150 mm $\times$ 2.1 mm, 3.5 μ M particle size) at 40°C with mobile phase A, 0.1% formic

acid in water, and mobile phase B, 0.1% formic acid in acetonitrile:methanol (40:60), at a flow rate of 0.2 mL/min. Analyst 1.7 was used for data acquisition and Sciex OS 3.4 was used for data processing and quantitation.

2.4 | Data Analysis

MassHunter Profinder V10.0 (Agilent) software was used to deconvolute data acquired in positive and negative ionization modes with retention time alignment. Two layers of putative metabolite identification were performed: (i) accurate-mass matching based on MS data alone (mass error $\leq$ 7 ppm) and (ii) MS/MS spectral matching against the METLIN MS/MS database using Mass Profiler Professional V15.1 (Agilent). When MS/MS data were insufficient to confidently distinguish between isobaric or closely related compounds, features were reported by molecular formula rather than by a specific metabolite name. Prior to statistical analysis, metabolite features with zero variance across samples were removed, as these do not contribute to group discrimination. Additional variance- or interquartile range-based filtering was not applied in order to avoid excluding potentially biologically relevant metabolites in this exploratory pilot study. Metabolites with missing values in more than 50% of samples were excluded. For the remaining features, missing values were imputed as 20% of the minimum detected value. The resulting peak tables were analyzed using MetaboAnalyst 6.0. Data were log-transformed and analyzed for statistical significance using a nonparametric two-tailed Mann-Whitney test ($p < 0.05$), with resulting p -values adjusted for multiple comparisons using the false discovery rate (FDR) method. Fold changes and adjusted p -values were integrated to generate volcano plots using the MetaboAnalyst platform. Random forest supervised machine learning was subsequently applied to classify samples as ADT-treated or eugonadal, and metabolites contributing most strongly to classification were visualized using variable importance plots. Finally, pathway analysis was performed on the MetaboAnalyst platform, based on the resulting important metabolites. To assess the correlation between serum and salivary glucocorticoids, a non-parametric rank correlation coefficient (Kendall's Tau) was utilized to measure the degree of concordance between the relative ranking of observations of each variable, and Figure 5 was prepared using SPSS 29 to depict this correlation.

Data Availability: The workflow of this study is depicted in Figure 1, and the deidentified data set is available in Supporting Information S1: File 1.

3 | Results

3.1 | Patient Characteristics

Baseline patient characteristics are listed in Table 1. Patients receiving ADT were more likely to have lower prostate-specific antigen (PSA) and higher Gleason scores. In serum and saliva samples across all patients, 1198 metabolites were identified. Initially, serum samples were compared to explore the differences between ADT patients and eugonadal patients. Next, saliva samples were compared to explore the salivary metabolomic profile of ADT samples vs. eugonadal patients, followed

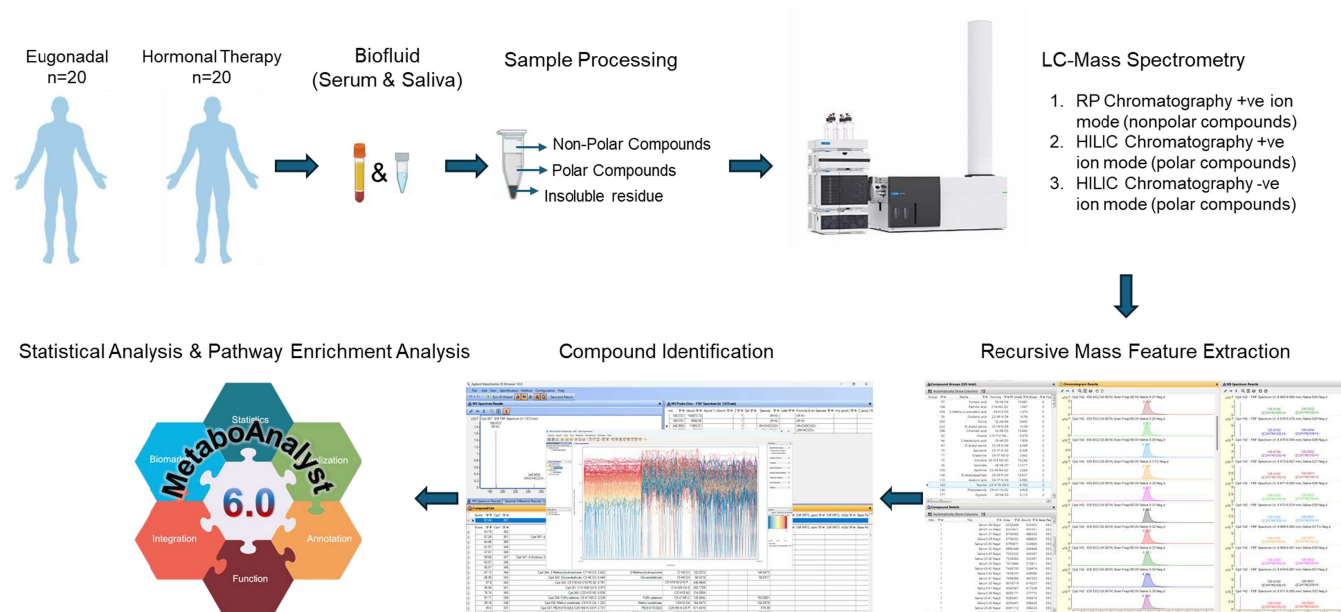


FIGURE 1 | Study overview and workflow for serum and saliva samples of prostate cancer patients recruited from University of Miami and Jackson Hospital from November 2023–July 2024. Matching blood and saliva were collected from 40 patients with prostate cancer: 20 eugonadal and 20 treated with hormonal therapy. Reversed-phase liquid chromatography was used to analyze nonpolar compounds whereas hydrophilic interaction liquid chromatography was the choice for polar compounds. Ion features were collected and identified for subsequent pathway enrichment analysis.

TABLE 1 | Characteristics of patients with prostate cancer according to androgen deprivation status.

Variable		Eugonadal (n = 20)	ADT (n = 20)	Total (n = 40)	p-value
Age (years)	Mean (SD)	67.4 (6.6)	67.7 (6.7)	67.5 (6.57)	0.82 Mann–Whitney
	Median (range)	66.0 (57–81)	66.5 (56–82)	66.0 (56–82)	
PSA (ng/mL)	Mean (SD)	7.3 (9.0)	0.5 (9.0)	3.9 (7.2)	< 0.001 Mann–Whitney
	Median (range)	4.41 (0.1–38)	0.01 (0.01–3.8)	0.21 (0.1–38)	
Stage	1C	2	1	3	1.00 Fisher's Exact
	2	1	1	2	
	3A	5	4	9	
	3B	4	4	8	
	4	5	6	11	
	Missing	3	4	7	
Gleason Score	3 + 3 and 3 + 4	9	3	12	0.06 χ^2
	4 + 3 and above	11	15	26	
	Missing	0	2	2	

by comparison of serum and saliva samples, using both untargeted and targeted methods for cortisol and cortisone.

3.2 | Comparison of ADT and Eugonadal Serum Demonstrated Differences in Amino Acids, Steroids, and Bile Acids Pathways

In the untargeted metabolomic analysis of serum samples, comparing ADT-treated patients to eugonadal patients revealed substantial metabolic differences. In patients receiving ADT, a total of 35 metabolites had a fold change (FC) > 2, indicating

upregulation, while another 35 metabolites had an FC < 0.5, indicating downregulation (Figure 2a). After applying the FDR-adjusted threshold of $p < 0.05$, 248 metabolites were found to be significantly different in ADT patients compared to eugonadal patients (Figure 2b). When integrating FC thresholds of > 2 or < 0.5 with an adjusted p -value of < 0.10, 14 were upregulated and 20 metabolites were downregulated in ADT patients compared to eugonadal patients (Figure 2c).

Random forest classification demonstrated a classification error rate of 2.5%, indicating robust discrimination between the two cohorts. The variable importance plot further highlighted key metabolites in classification of ADT or eugonadal samples (Figure 2d).

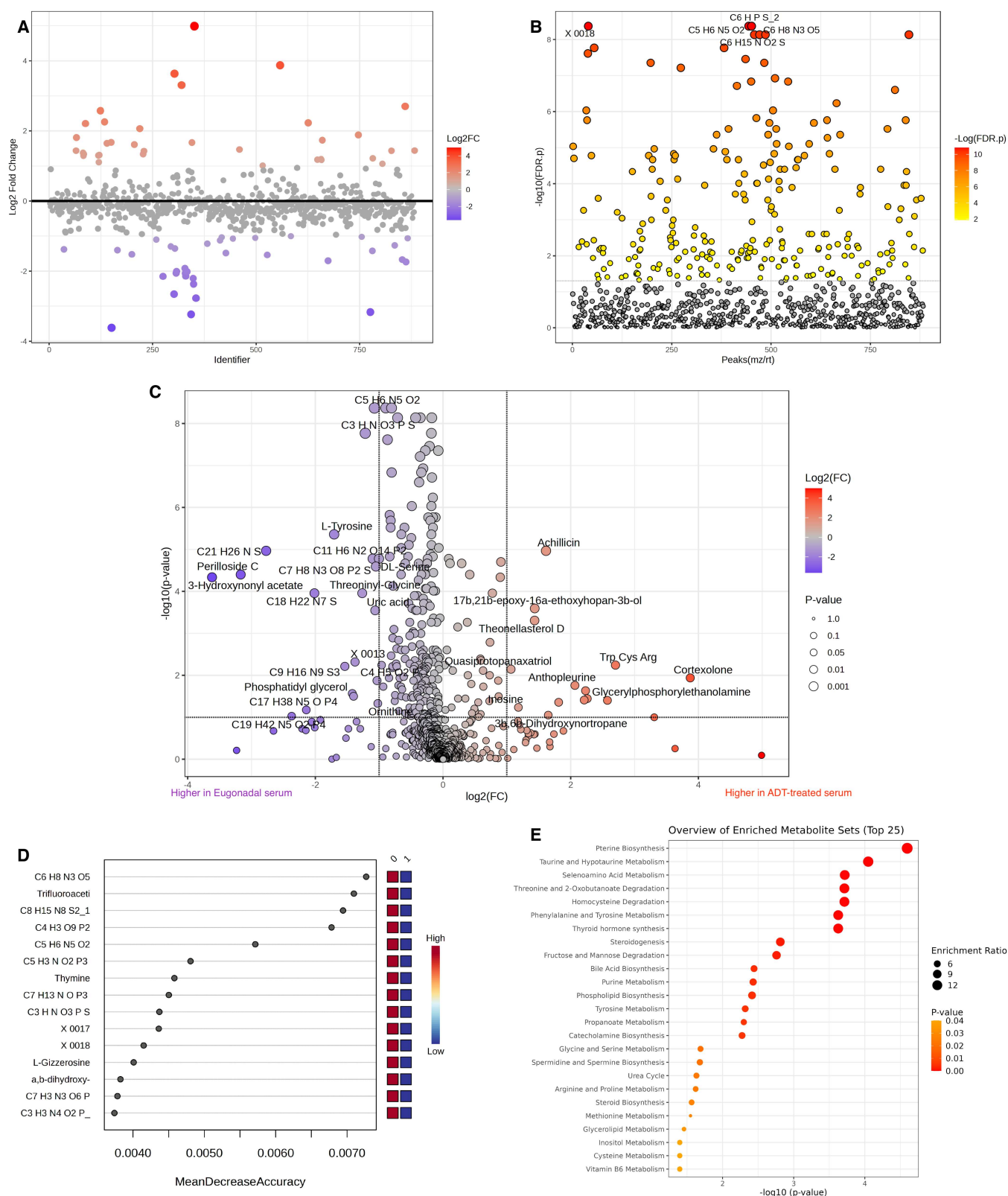


FIGURE 2 | Comparison of serum metabolites in eugonadal and ADT-treated prostate cancer patients, using untargeted LC/MS. (A) Log2 of fold changes of metabolites present in serum of patients with prostate cancer and ADT vs eugonadal patients, representing the proportion of metabolites with fold change > 2 (red) and fold change < 0.5 (blue). (B) -Log10 of p -values (adjusted using the false discovery rate) showing the proportion of serum metabolites that significantly differed (yellow and red) in eugonadal and ADT-treated patients. (C) Volcano plot of serum metabolites with significant differences (adjusted p -value > 0.1) and fold changes > 2 (red) or < 0.5 (blue) in ADT-treated patients vs eugonadal patients. (D) Variable importance plot of random forest supervised machine learning method to classify serum of eugonadal patients vs. ADT-treated patients; (E) Enrichment analysis plot demonstrating metabolite sets of different pathways different in serum of patients.

Enrichment analysis revealed important differences in key metabolic pathways, particularly in amino acid biosynthesis, steroidogenesis, and bile acid biosynthesis. The top 25 pathways that were affected are depicted in Figure 2e.

3.3 | Comparison of ADT and Eugonadal Saliva Metabolomic Profiles Demonstrated Differences in Amino Acids, Bile Acids, and Steroid Pathways

Analysis of saliva samples also showed significant and distinct metabolic alterations. In patients receiving ADT, 54 metabolites were upregulated ($FC > 2$), while 69 were downregulated ($FC < 0.5$) compared to eugonadal patients (Figure 3a). After FDR correction, 82 metabolites had statistically significant changes (adjusted p -value < 0.05 ; Figure 3b).

Using the same thresholds ($FC > 2$ or < 0.5 , $p < 0.1$), we generated a volcano plot, finding 28 metabolites to be downregulated, and 14 upregulated in patients receiving ADT (Figure 3c).

The random forest model for saliva-based classification achieved a 7.5% error rate, indicating moderate discriminatory power. The variable importance plot further highlighted key metabolites in classification of ADT vs. eugonadal samples (Figure 3d).

Enrichment analysis further confirmed disruptions in steroidogenesis, amino acid metabolism, bile acid synthesis, glutathione, and porphyrin metabolism. The top 25 pathways affected are depicted in Figure 3e.

3.4 | Paired Comparison of Untargeted Serum and Saliva Metabolites Showed Differences Attributable to Their Different Matrices

Untargeted metabolomics reports relative signal intensities that are strongly influenced by sample matrix, therefore direct comparisons between serum and saliva primarily reflect matrix-specific effects rather than biologically meaningful differences. Accordingly, paired comparisons were performed to characterize the extent of these matrix-driven differences prior to interpretation of ADT-associated metabolic changes.

A total of 538 metabolites were upregulated in saliva ($FC > 2$), while 408 were downregulated ($FC < 0.5$). After FDR correction, changes in 1128 metabolites were statistically significant (adjusted $p < 0.05$). Using thresholds of $FC > 2$ or < 0.5 with $p < 0.10$, 537 metabolites were upregulated, and 408 were downregulated in saliva compared to serum (Figure 4a).

In eugonadal participants, 535 metabolites were upregulated in saliva ($FC > 2$), while 404 were downregulated ($FC < 0.5$). Changes in a total of 1104 metabolites reached statistical significance after FDR correction (adjusted p -value < 0.05). Based on the FC and p -value criteria ($FC > 2$ or < 0.5 , $p < 0.10$), 537 metabolites were upregulated, and 379 were downregulated in saliva compared to serum (Figure 4b).

Among ADT-treated participants, 538 metabolites were upregulated in saliva ($FC > 2$), and 407 were downregulated ($FC < 0.5$). After FDR correction, 1106 metabolites were statistically significant (adjusted p -value < 0.05). Using $FC > 2$ or

< 0.5 with $p < 0.10$, 537 metabolites were upregulated, and 406 were downregulated in saliva compared to serum (Figure 4c).

3.5 | Serum and Saliva Demonstrated Higher Cortisol and Lower Cortisone Levels in Serum From Patients Receiving ADT, With Significant Correlation Between Serum and Saliva Concentrations

Considering that the differences above could be attributable to the matrix effect, and to better begin to understand the role of androgens in the salivary gland, we next used targeted metabolomic methods to compare serum and salivary cortisol levels.

The serum cortisol concentration in ADT patients was higher compared to eugonadal patients, although the difference was not significant ($p = 0.25$, Mann–Whitney U test). The concentration of cortisol in saliva of ADT patients was lower compared to eugonadal patients, although not statistically significant ($p = 0.18$, Mann–Whitney U test).

The concentration of cortisone in serum of ADT patients was lower compared to eugonadal patients, although not statistically significant ($p = 0.07$, Mann–Whitney U test). The concentration of cortisone in saliva of ADT patients was lower compared to eugonadal patients, although not statistically significant ($p = 0.13$, Mann–Whitney U test). Concentrations of cortisol and cortisone in serum and saliva of ADT and eugonadal patients are summarized in Table 2.

Among all participants, serum and salivary concentrations were moderately correlated for both cortisol and cortisone (cortisol, Kendall's Tau = 0.52, $p < 0.001$; cortisone, Kendall's tau = 0.71, $p < 0.001$). In ADT patients, serum and salivary cortisol had a correlation of 0.42 (Kendall's Tau, $p = 0.009$), and eugonadal patients had a correlation of 0.56 ($p = 0.002$). Finally, in ADT patients, serum and salivary cortisone were moderately correlated (Kendall's Tau, 0.53, $p = 0.003$) and somewhat more strongly in eugonadal patients, 0.65 ($p = 0.004$) (Figure 5a,b).

4 | Discussion

This study, through untargeted metabolomics of serum and saliva, provides a comprehensive analysis of metabolic alterations in prostate cancer patients who are eugonadal or undergoing ADT. The findings demonstrate systemic metabolic disruptions caused by ADT and highlight saliva as a promising, noninvasively obtained, biofluid for biomarker discovery.

The untargeted metabolomic analysis revealed significant disruptions in several metabolic pathways in serum samples from ADT-treated patients compared to eugonadal patients. The significant upregulation of 14 metabolites ($FC > 2$ and adjusted $p < 0.10$) and downregulation of another 20 ($FC < 0.5$ and adjusted $p < 0.10$) suggests differences in metabolomic profiles (Figure 2c). These findings align with prior studies demonstrating that ADT-induced androgen suppression profoundly impacts lipid and protein metabolism, including dysregulation of cholesterol and fatty acid synthesis [13–15]. Enrichment analysis of the metabolites also revealed significant changes in steroidogenesis, bile acid biosynthesis, thyroid hormone synthesis, and metabolism of many amino acids. Several other

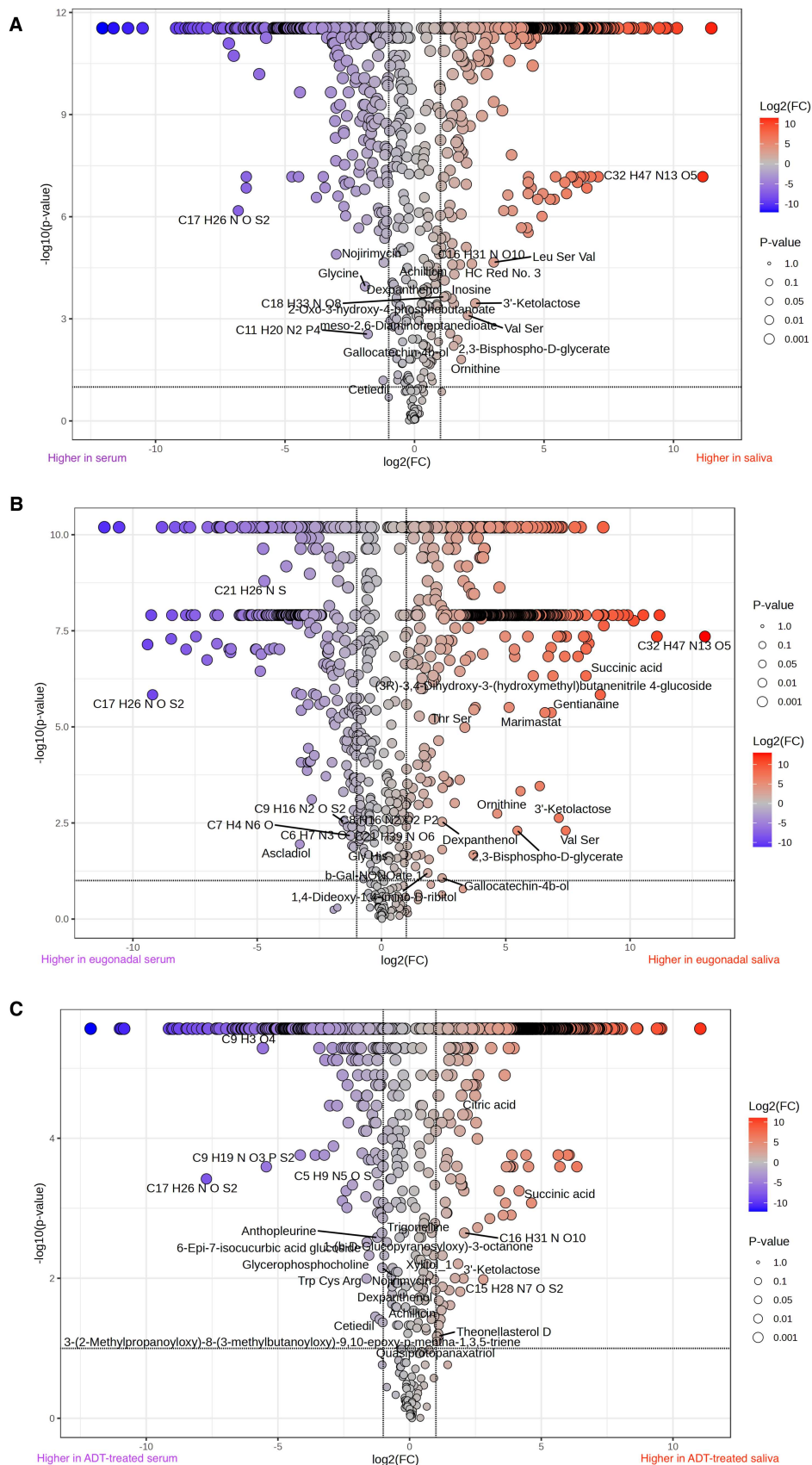


FIGURE 4 | Comparison of salivary enrichment of metabolites in eugonadal and ADT-treated prostate cancer patients, using untargeted LC/MS. (A) Volcano plot of serum vs. salivary metabolites with significant differences (adjusted p -value > 0.1) and fold change > 2 (red) or < 0.5 (blue) among all patients; (B) Volcano plot of serum vs. salivary metabolites with significant differences (adjusted p -value > 0.1) and fold change > 2 (red) or < 0.5 (blue) among eugonadal patients; (C) Volcano plot of serum vs. salivary metabolites with significant differences (adjusted p -value > 0.1) and fold change > 2 (red) or < 0.5 (blue) among ADT-treated patients.

TABLE 2 | Serum and salivary glucocorticoids in 40 prostate cancer patients based on their androgen deprivation therapy status.

Group	Cortisol (ng/mL)			Cortisone (ng/mL)		
	Serum		Saliva	Serum		Saliva
	Mean (SD)	Median (Range)	Mean (SD)	Mean (SD)	Median (Range)	Mean (SD)
Eugonadal	78.82 (24.92)	81.64 (24.47–132.11)	1.64 (1.02)	15.41 (3.93)	15.73 (7.36–22.49)	6.49 (5.31)
ADT	64.43 (55.41)	36.30 (3.68– 169.59)	1.48 (1.35)	9.19 (8.39)	4.46 (0.5–21.5)	5.16 (5.31)
All patients	72.21 (41.78)	74.96 (3.68– 169.59)	1.57 (1.17)	12.56 (7.02)	15.04 (0.5–22.49)	5.88 (4.19)
						5.08 (0.39– 17.89)

studies have pointed to similar changes induced by ADT. Bile acids have been shown to increase with ADT, raising the risk of bile stones; however, the mechanism is unclear [32]. Several studies have also demonstrated upregulation in steroidogenesis, similar to the findings in this study [33].

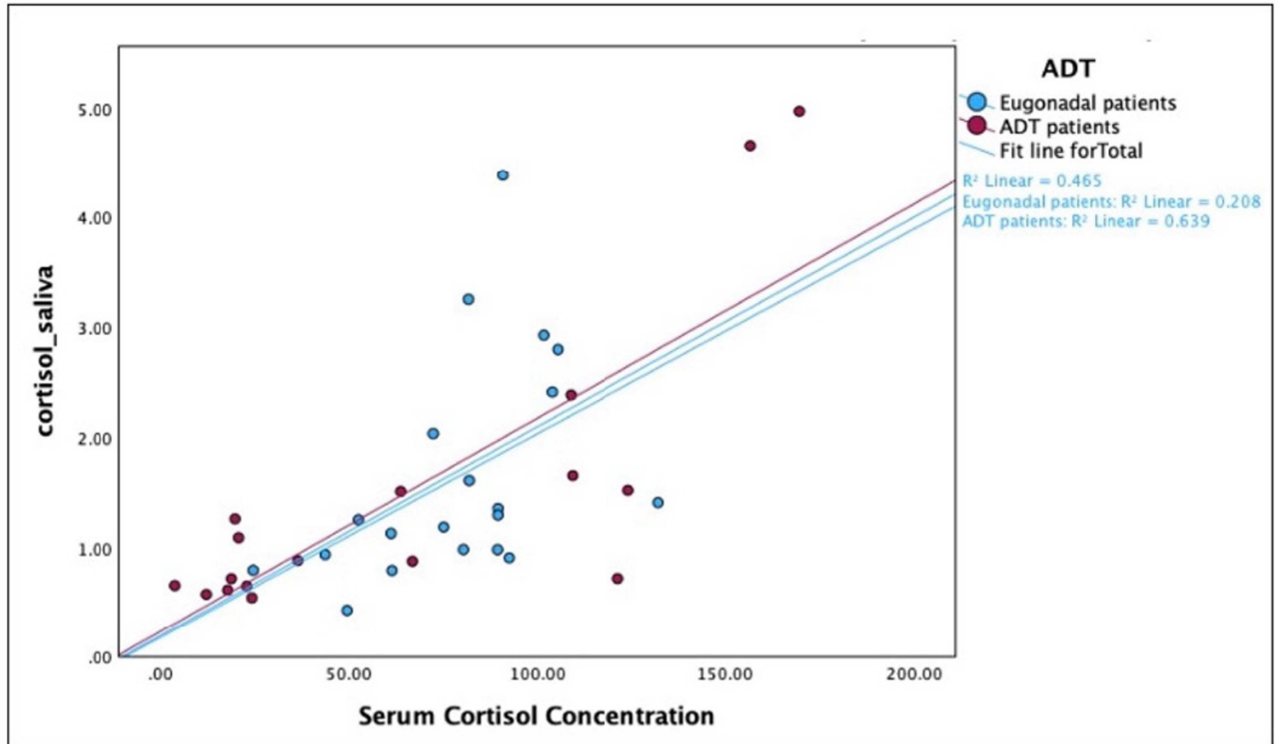
Saliva, as a biofluid, demonstrated potential as a biomarker of systemic metabolic changes in prostate cancer and treatment, as evidenced by significant alterations in salivary metabolomic profiles of ADT-treated patients. The relevance of salivary metabolites in monitoring ADT-induced metabolite changes is further reinforced by evidence suggesting that salivary gland function is directly influenced by androgen receptor activity. Several studies have demonstrated AR activity in salivary gland tumors, potentially highlighting the possibility of interaction of hormonal therapy with their physiology [17, 34–36]. These, along with our findings here, point to an important role of AR in salivary glands. 11 β -HSD2 has also been observed to be highly expressed in epithelial tissues such as salivary and colonic [37, 38]. Moreover, we have observed that in men, but not in women, AR is co-expressed with 11 β -HSD2 in the distal kidney tubules, and AR inhibition is associated with loss of renal 11 β -HSD2 activity [31]. The increased presence of androgen receptor activity in high-grade salivary gland tumors and the presence of significant metabolic changes associated with ADT suggest that AR activity plays an essential role in salivary gland physiology.

Our observations support the hypothesis that salivary metabolites and levels reflect those of serum due to local and systemic factors, possibly including enzymatic activity in salivary glands and endocrine influences from androgen suppression [7, 39]. The correlation between serum and salivary metabolomic profiles underscores the utility of saliva as a surrogate for serum in biomarker discovery. While serum metabolomics remains the gold standard for systemic profiling, saliva offers practical advantages, especially in community health settings where frequent serum collection for longitudinal studies may be challenging. Its noninvasive nature makes saliva a promising alternative, supporting further research into its integration for routine clinical monitoring, particularly for patients with limited access to healthcare facilities.

The ability of modern techniques to detect alterations in glucocorticoid metabolism in both serum and saliva is notable, and glucocorticoids play a central role in prostate cancer with regard to the stress response, metabolic regulation, and ADT-related metabolic disturbances [40–42]. In this study, serum and saliva cortisol and cortisone levels were highly and significantly correlated between eugonadal and ADT patients, which is in line with other observations reporting a similar correlation between serum and salivary glucocorticoids [10, 43–45]. Serum glucocorticoids are used in diagnosing adrenal conditions; however, the observed changes in cortisol and cortisone levels in both biofluids further validate the utility of saliva as a reliable matrix for assessing endocrine disruptions associated with ADT as well.

Future studies should explore the longitudinal impact of ADT on salivary metabolomic profiles and validate the identified biomarkers in larger, diverse cohorts. Additionally, targeted metabolomic approaches focusing on specific pathways, such as glucocorticoid metabolism pathways, could refine the utility of

A



B

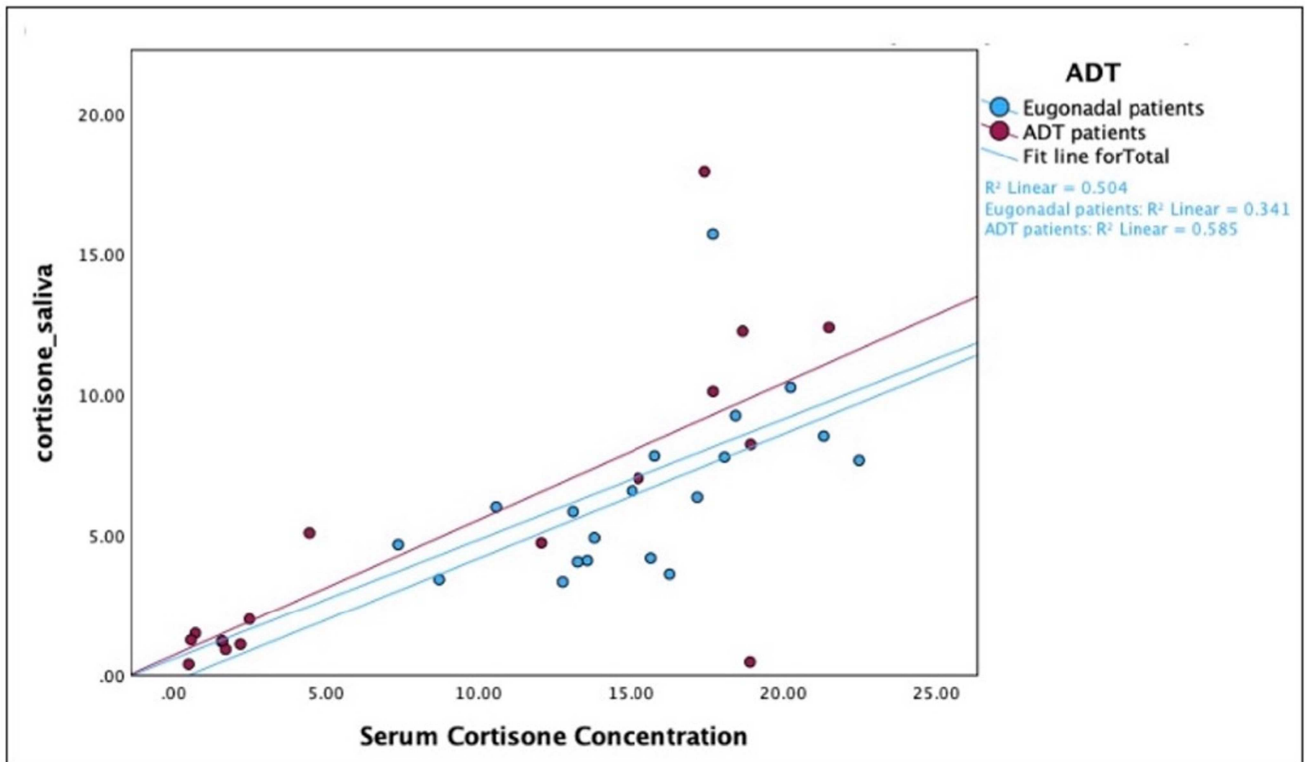


FIGURE 5 | Correlations between cortisol and cortisone in serum and saliva of participants. (A) Correlation between serum and salivary cortisol in eugonadal (blue) and ADT-treated (red) patients; (B) Correlation between serum and salivary cortisone in eugonadal (blue) and ADT-treated (red) patients.

salivary biomarkers for clinical decision-making. The development of standardized protocols for salivary metabolomics will also be crucial to ensure sustainability, reproducibility and generalizability in clinical settings.

Ultimately, the findings of this study hold important implications for the management of prostate cancer patients undergoing ADT. The integration of salivary metabolomics into clinical practice could facilitate early detection of metabolic alterations, enabling timely interventions to mitigate long-term health risks, such as cardiovascular disease, associated with ADT.

Integrating salivary metabolomics into future public health strategies can enhance community health by offering a non-invasive, cost-effective, and accessible method for sample collection. This approach improves patient monitoring and reduces participation barriers, particularly in underrepresented and underserved populations. The ease of saliva sampling also makes it an ideal tool for Community-Based Participatory Research (CBPR), fostering collaboration between community members and researchers to address local health concerns directly [5]. By empowering communities to take an active role in health research, saliva-based testing could improve health outcomes and contribute to more sustainable and tailored public health programs.

This study has several strengths and limitations. A key strength is the use of paired serum and saliva samples analyzed using both untargeted and targeted metabolomic approaches, allowing complementary assessment of global metabolic patterns and absolute quantification of selected endocrine metabolites. Untargeted metabolomics reports relative signal intensities rather than absolute metabolite concentrations, and direct cross-matrix comparisons of fold changes between serum and saliva are therefore confounded by matrix-specific effects. In contrast, targeted metabolomic analyses measuring absolute concentrations, as performed here for cortisol and cortisone, provide an appropriate framework for assessing the extent to which salivary metabolites reflect systemic metabolic changes.

Several limitations should also be acknowledged. The modest sample size and cross-sectional design limit generalizability and preclude assessment of longitudinal metabolic changes associated with androgen deprivation therapy. In addition, although all patients in the ADT group were actively receiving hormonal therapy at the time of sample collection, detailed information regarding duration of therapy and prior systemic treatments was not uniformly available for all participants, as some patients were referred from outside institutions. Accordingly, this work should be viewed as a proof-of-concept study demonstrating the feasibility of saliva as a surrogate biofluid for metabolomic profiling in prostate cancer.

5 | Conclusion

This study demonstrates the potential of integrating serum and salivary metabolomics to observe the metabolic effects of ADT in prostate cancer patients. The findings highlight the usefulness of salivary metabolomics as a non-invasive tool for biomarker discovery, treatment monitoring, and personalized care. By reducing gaps in healthcare accessibility and enabling earlier

detection of metabolic changes, salivary metabolomics holds promise for advancing precision medicine and addressing health disparities in prostate cancer care. Future studies should focus on the mechanisms underlying these findings and on exploring the translational potential of salivary metabolomics in clinical and public health settings.

Author Contributions

N.S., M.A., and A.A. conceived the project, designed experiments, interpreted data, and wrote the manuscript. M.A. performed most experiments. A.A. conducted statistical analysis. D.J.P., B.S., R.S., M.C.A., J.M., B.A.M., E.K., S.P., and M.B. provided the patient samples and L.P. helped with data interpretation. All authors have reviewed and approved the manuscript.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

Correspondence and material requests should be addressed to Nima Sharifi at nimasharifi@miami.edu. The deidentified data have been made available as Supporting Information S1: File 1.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section.

Final dataset to include as supplementary material_1.